Pepper Robot for Hospital Support
Elective in AI / HRI Report

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This report is only relevant to the part of HRI, 3 CFU (2023/2024 course).

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1 Introduction

1.1 Context and Motivation

The context of this interaction involves a robot operating within a hospital environment. Specifically, we use the humanoid robot Pepper, which will interact with patients by handling specific, yet impactful, tasks that enhance service quality while saving hospital staff valuable time. Pepper is responsible for managing patients' schedule reminders, such as following the therapeutic plan defined by doctors or scheduling relatives' meetings. With prior knowledge of schedules, the robot delivers clear and precise reminders, relieving staff from repetitive tasks that otherwise consume precious time. Simultaneously, Pepper addresses patients' emotional well-being by helping them cope with stressful moments or engaging them with interactive games. The goal is to positively influence their mood, regardless of patient type or condition.
In essence, Pepper functions as an autonomous hospital assistant. It can:

- Exploit past experience memory
- Adapts to individual users during mood regulation activities
- Autonomously requests human intervention at the patient's request
- Takes the initiative to propose activities to the user

ensuring comprehensive and safe support in a patient-centred care experience. During interactions, we assume that Pepper is placed in front of the patient to provide its services. In this task, managing the patient's emotional state, providing entertainment, and remembering schedules are the primary objectives.

1.2 Motivation and Benefits in Hospital Context

Implementing and utilizing this tool within a hospital setting can generate multiple positive outcomes socially, operationally, and economically. In healthcare environments, where stress adversely affects the well-being of patients, adopting innovative solutions is critical. We use Pepper, a humanoid robot developed by SoftBank Robotics [3], which, in our case, is designed to provide emotional support through guided relaxation exercises centred on controlled breathing cycles aimed at reducing anxiety and promoting calmness in patients. In addition to relaxation routines, Pepper can also engage patients with light entertainment, specifically jokes and riddles, further contributing to a more pleasant, emotionally supportive and stimulating environment [6]. Finally, it also provides some nursing support, such as helping patients stand up or remembering their schedules.
Thanks to its ability to interact empathetically and adaptively, Pepper can lead breathing exercises or propose funny entertainment that ease anxiety and enhance the patient's experience during hospitalization. Indeed, researches [13], [7], show that interactions with social robots like Pepper can positively influence

patients' emotional states, contributing to a more welcoming and soothing care environment.
Patient stress is often underestimated in hospitals. Although, numerous studies have highlighted how stress and anxiety negatively impact recovery [15] [20], clinical practice frequently overlooks these factors. Multiple reasons underlie this oversight: traditional hospital organization prioritizes physical disease treatment (diagnosis and interventions), leading to less focus on patients' psychological and emotional needs; number of workers in a hospital and limited resources impede adequate attention to mental well-being.
It is important to emphasize that addressing patient stress not only improves their hospital experience but can also accelerate recovery and reduce the length of stay. Furthermore, deploying Pepper frees medical staff, allowing them to focus on more complex clinical duties by delegating emotional support and mood monitoring to the robot. This integrated approach enhances healthcare system efficiency and fosters patient-centred care.
In summary, using Pepper assistance in a hospital offers a dual benefit: it improves patients' psychological well-being and optimizes staff workload, marking an advancement toward more human-centred, tech-enhanced healthcare.
From an economic standpoint, Pepper helps optimize human resources by taking on repetitive support tasks, freeing up medical staff for higher-value responsibilities. This task redistribution can increase operational efficiency and reduce costs associated with stress management and burnout, especially crucial given recent staffing shortages. Over the last 11 years, Italy's National Health Service lost over €28 billion and saw a significant decline in healthcare personnel [8, 12], including doctors and nurses.
This factor has increased workloads for the remaining staff, resulting in heightened stress and burnout risk. Introducing robots like Pepper helps mitigate these issues by covering support roles, ranging from basic care reminders to stress management and patient engagement.

1.3 Objectives

In this HRI task, our objective is to demonstrate a social robot's ability, specifically Pepper's, to recognize and manage the signs of stress and boredom in patients, and deal with simple nursing tasks. This is done by responding appropriately through guided relaxation exercises, entertainment activities, and providing simple health-care support. This involves adapting the interaction to individual user preferences, considering also past history. Pepper encourages users using tailored reassuring phrases, moves appropriately, engages in consensual physical contact contextually to the interaction, and takes initiative when possible, all delivered through voice, screen display, and motion. We aim to test how effectively a personalized human–robot interaction can promote the emotional well-being of hospitalized patients. Through this project, we explore how integrating social robots can humanize healthcare, provide emotional support, and alleviate the workload of medical staff. Our goal is to deliver concrete evidence on how emotionally supportive behaviour, implemented via voice, screen,

and movement can improve patient well-being and ease the burden on healthcare workers.

1.4 Summary of the Results

We have shown that the robot can be deployed in a hospital setting, is able to remember users, their medication schedules, and exercise preferences, while also providing interactive engagement. During stressful moments, the robot successfully guides the user, encourages them, adapts its behaviour, and performs appropriate movements. The robot shows healthcare abilities; for example, helping a patient to reach a standing position and monitoring their schedule. Additionally, Pepper demonstrates initiative by suggesting entertaining activities such as telling a joke or asking a riddle.

2 Related Work

Much of the inspiration for the functionalities implemented in this project comes directly from the course lectures.
Introduction slides [16] were useful for determining the level of autonomy we wanted to give the robot, and for considering how to resolve a task in a socially acceptable manner. In this project, we want to achieve a task by paying attention to the words spoken and to how the robot interacts with the user.
The slides on social signals [17] were fundamental to constructing an interaction considering social signals in input and output, and taking care to include in our project also non-verbal signals, which are fundamental for effective communication. In fact our robot is able to express feelings using various types of signals. The handling of social signals has been a central focus: both their perception and generation, as well as the reasoning processes based on these cues. Reasoning about these social signals to consider various scenarios was also crucial in defining some of the interactions, in particular, reasoning beyond purely reactive responses.
The contribution of the slides on Multi-modal interaction [18] is the basis of our interaction; it inspired us to achieve interaction based on multiple communication channels. The interaction was designed to be inherently multimodal, combining voice, gestures, screen display, and tactile input to enrich communication between the robot and the user.
From the Motion control slides [19], we got the idea that motion control is the fundamental distinguishing element in HRI: whereas in classic human-computer interaction, communication takes place mainly on visual and textual channels, here, the body and physical contact become active information carriers. Movement must convey intentions, feedback, and safety. Any gesture or movement of the robot must be designed to communicate the internal state and future actions, always integrating with at least one other channel (e.g. display).
During lectures, the topic of memory was also central, which is why we focused on the integration of a memory of past events and prior knowledge about the

environment, so that the system not only reacts to current stimuli but can evaluate contexts, anticipate situations and modulate its responses on the basis of accumulated experience, overcoming the "if-then" paradigm.
Additionally, this work was also inspired by the "Ethics of AI" seminar [10], where the topic of trustworthiness was discussed in depth. Trust is a fundamental element for ensuring the social acceptability of autonomous systems. To support this, the robot was designed to clearly communicate its capabilities and limitations to the user, thereby reducing perceived uncertainty and increasing confidence in the interaction.
In healthcare facilities, the need for assistive and interactive robots has increased, as standards rise while the number of staff members remains limited. These robots are designed to support patients alongside human nurses and other members of the medical staff. One such example is RONA (Robot Nurse Rona) [5], a robotic nurse developed with advanced facial recognition capabilities. It can operate continuously for up to six hours and perform medical measurements to reduce the workload on human nurses. However, beyond physical assistance, there is a growing need for social-cognitive robots that can engage with patients and provide emotional support and social interaction—these are known as Socially Assistive Robots. In this context, the Pepper robot was employed in a study [11] conducted in an institutional care facility, where it was found that patients enjoyed interacting with the robot. ROBEAR is another experimental robot designed to perform various tasks, including safely lifting patients and assisting individuals who need support to stand. This robot was developed with careful attention to patients' social responses. Its appearance resembles a friendly bear, specifically to avoid the well-known "uncanny valley" effect.
In our project, we consider a general patient without age restrictions. A study [22] demonstrated the efficacy of the interaction between children and the Nao social robot in reducing stress levels, particularly in terms of decreased cortisol levels. This systematic review [14], which is central to the choice of social robot for our work, shows how social robots in elderly people without cognitive impairment and with dementia, help in a positive way, providing security, and reducing stress.

3 Solution

3.1 Architecture

The core component of the system is the **NAOqi Server** [2], which is the main software that runs on the robot and controls it. In our case, it simulates the robot's hardware and provides a suite of modules for interacting with the robot. These modules, called **services**, give a direct interface to the robot sensors (e.g., touch sensors on the hands), actuators (e.g., robot limbs and head), and some processing components (e.g. automatic speech recognition system). The NAOqi services involved in our system are:

- **ALMemory**: manages the central memory of the robot, used to store

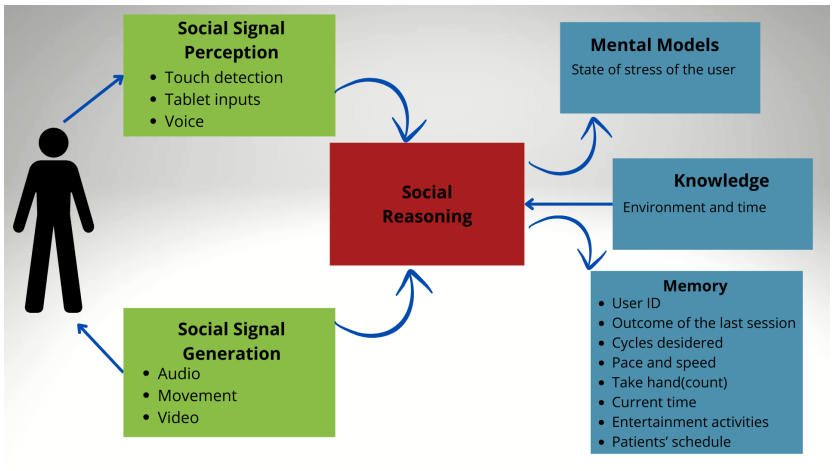


Figure 1: The architecture design

all key information related to the robot's hardware configuration and its current representation of the state of the world (e.g., information about the user it is interacting with).

- **ALDialog**: manages the different steps of the communication interactions with the robot
- **ALTouch**: handles the interactions with the touch sensors of the robot
- **ALMotion**: handles the movements of all the joints of the robot and eventually the movement of the whole chassis

Another key component of the system is the **MODIM** node [1], which supports **multi-modal** tablet interactions via the MODIM framework. MODIM consists of two components, a server and a client. In our system, both components run locally (the client is served as a web page), although in a real deployment, they would typically run on separate machines. Multi-modality is a key component for social robots, and especially in a hospital environment, we consider it even more important not to limit the interactions to a single modality, providing a better user experience, improving usability and accessibility.
We designed multi-modal interactions involving the following modalities:

- **Tablet Visualization**: images, text, interactive buttons
- **Text-to-Speak** Robot speech output
- **ASR** Audio-speech recognition

A key aspect of a cognitive-social robot is the ability to exploit past interactions to provide a more **personalized user experience**. To this purpose, we

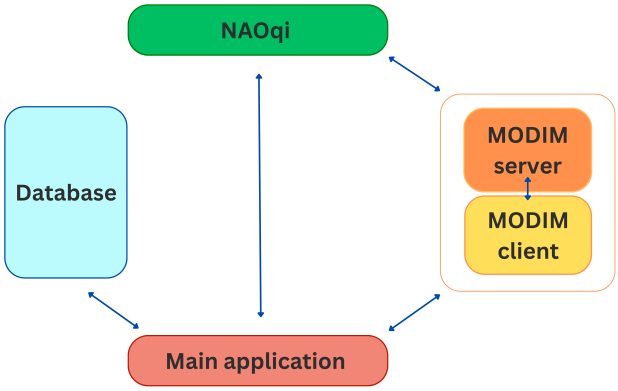


Figure 2: The architecture: shows the modules of the system and how they interact with each other

utilized a small **database** where persistent data about the users' past interactions and preferences are stored.
Finally, a central **Main** node orchestrates all the mentioned components (see Figure 2), accessing the NAOqi services, initiating tablet interactions, loading, saving, and processing user data from the database, and defining the logic of the order of the interactions executed.

3.2 Welcome and User Management

The system comprises an initial phase where the robot **establishes contact** with a user, followed by a second phase in which the robot and the user engage in **specific interactions** provided by the robot. These interactions may take place verbally or through the tablet, sometimes both together (multi-modal interactions). The robot may execute some animations contextual to the situation to express emotions or to provide physical support.
The whole interaction starts with the robot asking about the generalities of the user it is interacting with, precisely their first and last name. If the user was registered before in the database, the robot loads their data into the central memory and exploits them for personalizing the following interactions, eventually updating them (for example, when it receives some feedback after the stress management session). If the user is interacting with the robot for the first time, they are prompted to register with the robot database.
The robot proposes the registration, expressing the advantage of exploiting the history of the previous interactions for a **personalized experience**. This possibility is always offered as a proposal, and the user is never forced to register in the database. This may be the case when a user wants to keep their information

private or wants to skip the registration phase for a quick interaction. If the user decides not to register, their information will not be persistent and will be overwritten when the robot starts to interact with a new user.
When agreed, the registration consists of a series of questions. Each question is followed up by a confirmation of the data to insert; this helps avoid errors and gives the user full transparency.
After dealing with user registration (potentially, the user is a guest user), the robot presents a **clear** and **explicit** list of its abilities to reduce uncertainty and increase confidence in the interaction [10]. The user can ask for clarification and eventually start a specific interaction. Sometimes the robot may propose to engage in an interaction, based on some **social cues** [23]: for example, if the user expresses some stress, by stating it out loud, the robot may propose to deal with it in a stress management session.

3.3 Nursing

The robot provides some nursing interactions designed with the core idea of helping the patient and reducing the workload on human nurses; however, the robot must be intended only as an assistant, and it should never be considered as a replacement for a professional figure.
The main interaction of this section is the **schedule interaction** in which the patient asks the robot to show their schedule. This is only possible if the user is registered in the database; otherwise, the robot will not have access to this information. The interaction involves the use of the tablet. The patient can choose between their daily schedule, the weekly one, or the full schedule. This can be done either by speaking or pressing the buttons presented on the monitor of the tablet. The resulting schedule is shown on the tablet as a list of events with their dates, times, and names of the event, which can be, for example, a reminder of a certain pill or medication the patient requires.
The system presents some **emergency** interactions that can always be accessed by the patient's request. The first of these is the call for human assistance; at any moment, the patient can interrupt the current interaction and ask for human help. The robot will instantly ask for confirmation if medical attention is required and proceed to submit the request.
Furthermore, the patient can ask at any moment for help standing up. In this case, the robot will assume a helping position and propose to touch its hands until necessary. The robot detects a touch, and when a release is detected, it asks if help is still required. In case no more help is necessary, it will assume a base standing position.

3.4 Stress Management

Here we talk about the relaxation exercise interaction.
One of the specific interactions Pepper is capable of performing within the hospital environment is a relaxation exercise, based on cycles of inhalation and exhalation, designed to calm the user. This proves useful in managing typical

stress that can arise in such settings.
Pepper can either autonomously initiate the interaction by interpreting certain keywords in the user's speech or respond to an explicit request. This is possible because Pepper can process audio input and understand what the patient is saying, reacting accordingly.
If the user expresses stress during a conversation, for instance by saying "I feel stressed," Pepper takes the initiative and suggests the relaxation exercise, which must then be confirmed before it begins. Alternatively, if the user explicitly says he/She want to do a stress exercise, the robot asks for confirmation and then starts the activity.
Once the robot has understood that the interaction should begin, it recognizes the user and adapts its behaviour.
For instance, if the user is not new and has previously provided feedback, using its memory, Pepper will personalize the exercise by modifying the number of breathing cycles, adjusting the pace of its voice, and altering its movements, such as extending its hands toward the user, but also changing the opening and closing dialogues.
For example, if the user wants to increase the number of cycles, and wants a slower pace, pepper for the next interaction will do more breathing cycles, and speak in a slower pace. All this is stored in memory, which is then used to reason about it and adapt.
This memory is modifiable at the user's choice, where at the end of the exercise, the user is asked if he/she wants to leave feedback, if so, he/she will be asked a series of questions to better customise the interaction.
These variables change only if the user has submitted feedback; otherwise, default settings are used, which are also applied to new users. Remembering what happened in previous sessions is essential for ensuring continuity and giving the user the impression of being recognized. About what we memorize, see the next section.
In the case of past negative interactions, Pepper still maintains a positive and encouraging attitude. It motivates the user to try again, relying on previous feedback where available. The goal is not simply to complete the task, but to do so in a socially acceptable way using supportive language that builds trust and encourages the user to retry, while reassuring them that improvements have been made.
So the behaviour changes depending on whether the previous interaction was positive or negative, changing how the speech begins at the new interaction.
The robot will speak and move differently, not due to an explicit command, but based on the outcome and interaction history, since the system is based on the use of memory, in which we store feedback and past experiences , pepper will reason on them, as said before, if the outcome of the session is positive, then it shows happiness, otherwise if negative, sadness, but with hope for better results by adapting the exercise.
Pepper thus reflects on past experiences to structure the interaction differently and make it more dynamic.
Movement plays a fundamental role in this exercise. During the breathing pro-

cess, Pepper guides the user by raising and lowering its head and torso to encourage inhalation and exhalation. It also performs gestures at the beginning of the exercise, both to associate a specific action with relaxation exercise (as an anchor gesture), and to emphasize the emotions it intends to convey, such as joy or sadness, enhancing the emotional impact.
The only movement explicitly triggered by the user is the act of offering its hands, in case the patient wants to hold them during the exercise, but if the user has always accepted or always refused to take the hand in the last interactions, pepper avoids asking the question and performs what the user usually chooses.
At the core of this interaction is the breathing exercise itself, proposed as a tool to help a person calm down during a stressful moment. The exercise begins with a short introduction that helps the person focus on their body and breathing, and on managing intrusive thoughts.
Pepper then alternates between inhalation and exhalation phases, each lasting five seconds.
This inhale–exhale cycle can be repeated several times. At the end, the session can either be concluded with a farewell, restarted, or ask to call a nurse, if the session does not proceed as expected, who will take over if Pepper is unable to effectively handle the stress.
Also in the final part the user can change Pepper's setting in the feedback part to help Pepper better adapt the experience to their preferences in the future.
Feedback can be given in two ways: by speaking directly to Pepper or by pressing buttons on the tablet display. We use feedback to personalize interaction, enhancing the sense of familiarity and comfort, two key elements for the acceptability of robot-assisted interactions.
We also worked on the interaction Robustness. For certain questions, Pepper does not rely solely on voice input but also accepts responses via touch on the tablet. This dual-input approach enhances reliability from the robot's perspective while simultaneously displaying available options on the screen, which helps prevent misunderstandings and ensures the human does not miss any information.
A key function of the tablet interface is to narrow down the range of possible user responses, reducing ambiguity. For example, even if a user verbally expresses the desire to engage in a particular interaction, Pepper always asks for confirmation before proceeding to avoid accidental triggers.
Pepper also shows context-awareness, since we assume a function that allows Pepper to access the current time. With this, the robot can autonomously lower its voice volume during evening hours, complying with hospital standards that require reduced noise levels to ensure patient rest. This contextual sensitivity improves Pepper's acceptability among both patients and medical staff.
About the Autonomy Level, the level of autonomy is variable: Pepper can initiate the suggestion to begin the relaxation session, but it will only proceed if the patient explicitly consents.
In this part of the exercise, the monitor is also used a lot, showing a calming blue light during the breathing cycles, showing whether to inhale or exhale, and

the number of cycles performed.

3.5 Entertainment

The last type of activity that patients can engage in with Pepper is focused on entertainment. This functionality has been designed to improve patients' experience by offering interactive and enjoyable moments that go beyond basic utility or assistance. When selecting this activity, patients are presented with two distinct modes of interaction, each varying in complexity and cognitive engagement.
The first mode is relatively simple and consists of Pepper telling **jokes**. This interaction aims to generate smiles and laughter, favouring a sense of companionship and well-being. It is particularly valuable in settings involving elderly users, where moments of light-heartedness can have a significant positive impact on mood and emotional health. Humour, even in short exchanges, has been shown to relieve stress, reduce feelings of isolation, and contribute to an overall improvement in mental state.
Below is an example of a joke

```
What do you do to have a space party? You planet!
```

The second mode is more cognitively challenging and consists of multiple-choice **riddles** posed by Pepper. This interactive format encourages the user to think critically, reflect on possible answers, and engage in problem-solving. Unlike passive forms of entertainment, riddles stimulate mental activity and can help maintain cognitive functions such as memory, reasoning, and creativity. For elderly people, particularly those living in environments with limited stimulation, this type of engagement can play an important role in delaying cognitive decline and preserving mental sharpness.
Following is an example of a riddle

```
I'm full of keys but I can't open any door. What am I?
```

A. `A Piano`

B. `A Keychain`

C. `A Safe`

D. `A Jail Cell`

E. `A Computer Keyboard`

These entertainment interactions are not merely recreational but serve an essential therapeutic purpose. In the context of elderly care, it is well understood that psychological stimulation is just as important as physical well-being. Pepper's ability to provide both humour and intellectual challenge addresses this need, helping to combat loneliness, boredom, and the mental stagnation that

can accompany ageing. As such, the entertainment activities offered by Pepper are carefully crafted to support overall well-being through a combination of emotional uplift and cognitive engagement.

3.6 Summary

The solution is based on a modular architecture in which a Main node coordinates NAOqi services for memory, dialogue, touch and movement management together with the MODIM framework for multimodal interactions via tablet, and supports three integrated functional areas: user reception and management with optional registration and personalisation based on the memory of previous interactions; nursing support, which allows users to view their treatment plan, request medical assistance and receive help getting up; and finally, wellness interactions, which include adaptive relaxation exercises synchronised with the user's movements and feedback, as well as entertainment such as jokes and riddles to stimulate mood and cognitive abilities.
In order to make the key concepts clearer, below we show what the robot produces, analyses and how it reasons for each of its possible objectives.

3.6.1 Reasoning Module

Upon recognizing a returning user, Pepper consults its memory to adapt its behaviour. Rather than reacting blindly or in a fully reactive way to current stimuli, the robot adjusts both its verbal and non-verbal responses, drawing on past interactions to shape its current behaviour. This includes:

- Remembering past feedback to adjust encouragement or timing.
- Adapting its initial statements based on how previous sessions went.
- Conclude differently depending on how the current session went.
- Deciding whether to offer its hand.
- Propose exercises according to the user's needs, either on their own initiative or explicitly requested.

This reflects the implementation of *social reasoning*, one of the most essential aspects of this interaction module. It allows Pepper to reason beyond surface-level inputs, behaving more like a socially intelligent companion than a reactive machine.

3.6.2 Social Cues Considered for Pepper

- **Verbal Cues**
  - Speech rate
  - Strategic pauses

Both adapted to project either a calming or more dynamic interaction style.

- **Non-Verbal Visual Cues**
  - Gestures and postures (e.g., synchronized torso/head movement with breathing cycles)
  - Tablet animations (e.g., full-screen blue light during relaxation, displaying options for feedback, images)
- **Tactile Cues**
  - Hand contact as a comforting gesture
  - Hand contact to establish the beginning and the end of the "help standing" interaction
  - Tablet input for robust communication

- **Motion-Based Signals** The body, as a multimodal output channel, is essential in stress-reduction exercises and represents a key element that distinguishes HRI tasks. For example, when a relaxation session ends successfully, Pepper expresses joy not only through voice but also by physically raising its arms.
  - Movements to associate the start of the exercise, but also to express joy or sadness
  - Reaching a helping pose to assist the patient in reaching a standing stance
  - Motion to express different type of exercise

3.6.3 User Input Considered

Verbal Input: Pepper interprets spontaneous verbal expressions of stress (e.g., "I feel stressed," "I'm anxious") as triggers to propose the relaxation exercise. Additionally, explicit user requests, such as "I want to do a relaxation exercise," directly prompt the robot to offer the activity. Verbal feedback at the end of the exercise ("I feel better," "it was too fast," etc.) allows the robot to adjust future interactions accordingly. Pepper also responds to simple confirmation phrases ("Yes, I want to start the exercise") and can optionally accept vocal commands to modify interaction parameters ("slower," "more cycles," etc.). Users can also verbally request light entertainment, such as jokes or riddles; to do it, sentences like "I am feeling uninspired; what can I do to pass time?", "Could you please tell me a riddle?" or "Could you please share a joke with me?" can be used.
Touch Input (Tablet): The tablet allows users to specify their preferences regarding the number of breathing cycles (increase/decrease), adjust speech speed, and manage pauses. At the conclusion of an exercise, predefined feedback options can be selected to refine subsequent sessions. Furthermore, tablet input

serves as an alternative communication method for users who have difficulties providing verbal input, enhancing the robustness of the interaction. As a complementary function, the tablet can display short riddles for entertainment and can be used to select the chosen answer by pressing the associated button.
Session History-Based Input (Memory-Driven): Information collected from previous interactions are used to guide future interactions.

4 Implementation

4.1 Main Node

The system is entirely simulated in a local environment on the Ubuntu distribution, but it can easily run on a real robot thanks to its modular design. To visualize and interact with the simulated robot, we utilized Choregraphe [21]. Some other interactions, like the touch of a touch sensor, are simulated, inserting a certain value over the required sensor measurement using the NAOqi Python SDK .
We utilized **NAOqi Python SDK** to have direct access to the modules that run on the robot system. We also utilized **Pepper Tools** [4] to have a higher level of access to the same modules.
The **Main** program uses the NAOqi Python SDK to access these modules in simulation, invoke predefined functions and events, and register custom callback functions to trigger under specific conditions. We also use the Pepper Tools interface, which offers a higher-level Robot class with simple, easy-to-use methods for posture control, touch feedback, and access to NAOqi's shared memory.
The main node starts up the connection to all the NAOqi modules necessary and starts a callback loop, where the logic of the robot is designed in response to the interaction it is having with the user. For example, if a user asks to visualize their daily schedule, a callback on the main node will be called, which will stop any other interactions and start the schedule interaction over the tablet application.

4.2 Dialogues

An important service we employ is **ALDialog**. We use it to design all spoken interactions between the human and the robot (except when the tablet is involved). ALDialog works by defining a set of topics that can be activated or deactivated depending on the current phase of the conversation. Each topic is defined in a different file, and more than one topic can be activated at the same time. Each topic contains possible user inputs and robot prompts, which can be triggered by other topics or by scripts.
To make interactions more natural, each concept can be expressed in multiple ways. For instance, if the robot asks a yes/no question, an affirmative reply might be "yes," "sure," "yeah," or any similar phrasing (Figure 3). Likewise, the

robot can rephrase its prompts without changing their meaning. This flexibility enables more engaging conversations and frees users from having to memorize exact trigger phrases. We exploit a pre-made topic file, called **lexicon_enu**, which contains many English concepts expressed in multiple ways; for example, the request for help is phrased in many different ways.

```
# User asks directly for a riddle
u:(["I would like you to give me a riddle."
    "Could you please tell me a riddle?"
    "I'd appreciate it if you could offer me a riddle."])
%startRiddle  Here's a riddle for you
```

Figure 3: A basic interaction designed with ALDialog: the user can propose to start the interaction in multiple ways. The robot responds, and the tag **%startRiddle** is caught by the Main program, which starts the tablet interaction.

Each dialogue can branch and conclude differently based on the user's requests and responses. For example, if the user asks for entertainment, the robot may offer a riddle. If the user responds affirmatively, the robot will launch a riddle interaction on the tablet; otherwise, it will suggest another entertainment option, such as telling a joke. Because tablet-based interactions bypass ALDialog and instead use NAOqi's standard Audio Speech Recognition (ASR) module, we disable all ALDialog topics whenever a tablet session begins to prevent conflicts between the two channels. The sole exception is the Emergency topic, which remains active at all times so users can always issue an emergency request (Figure **??**).

4.3 Choregraphe

Choregraphe is our primary tool for **visualizing** and **interacting** with the robot (Figure 4). It provides a visual simulation of a virtual robot that mirrors the real robot's posture, movements, and spoken output, and it includes a chat box directly connected to ALDialog, which allows writing the user inputs that will be received by the robot as if a human were speaking in front of it. By default, Choregraphe's chat box does not allow typing inputs for the ASR NAOqi module (only the ALDialog channel). For this reason, we implemented a callback function that intercepts every spoken sentence on the ALDialog and replays it into the NAOqi ASR module. This simple trick lets us use the same chat interface for both ALDialog and tablet-driven dialogues.

4.4 Tablet

The system includes a series of multi-modal interactions that involve the tablet. For this purpose, we utilized the **MODIM** framework. With it, we defined three

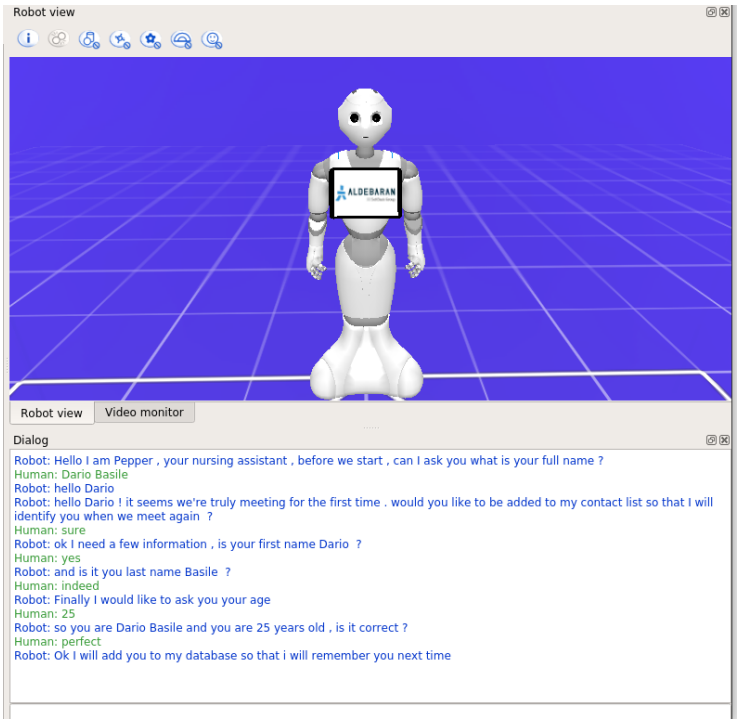


Figure 4: Choregraphe visualization of the robot and the Dialog chat-box

main interactions, which are: the **stress management session interactions**, **schedule interaction**, **riddle interaction**.
When an interaction gets started by the Main node, the ALDialog topics get deactivated, except for the emergency topic, which is always available to the user. Each interaction is defined as a function within a Python file, each of them containing the flow of the interaction along with the call to many **multi-modal actions**. These actions involve multiple modalities of input and output, combining the spoken dialogues with the tablet interface.
Each action is described in a configuration file that specifies which modalities to use and their parameters. Figure 5 shows an action file with many of the most common modalities utilized.

Many of these actions expect a response from the user, which is used in the Python interaction file to decide the direction in which the interaction continues. When the current interaction terminates, the Main node proceeds to re-activate all the ALDialog topics that were deactivated before.

4.5 Database

The database is implemented using **SQLite** [9], which is a database management system based on SQL, which works locally. The whole database is contained in

```
IMAGE
<*,*,*,*>: img/riddle_action.jpg
----
TEXT
<*,*,*,*>:  I’m full of keys but I can’t open any door. What am I?
----
TTS
<*,*,*,*>: I’m full of keys but I can’t open any door. What am I?
----
BUTTONS
wrong
<*,*,*,*>:  A. A Safe
wrong
<*,*,*,*>:  B. A Keychain
correct
<*,*,*,*>:  C. A Piano
wrong
<*,*,*,*>:  D. A Jail Cell
wrong
<*,*,*,*>:  E. A Computer Keyboard
----
ASR
<*,*,*,*>:  { 'correct': ['C', 'c', 'third', '3', 'A Piano'],
              'wrong': ['A', 'B', 'D', 'E', 'a', 'b', 'd', 'e', 'first', 'second', 'fourth', 'fifth',
                        '1', '2', '4', '5', 'A Safe', 'A Keychain', 'A Jail Cell', 'A Computer Keyboard'] }
----
```

Figure 5: The tablet action that proposes a riddle to the user: the action includes an image for context to the riddle, the riddle itself, written on the tablet and spoken by the robot, 5 buttons with possible solutions, of which only one is correct, and finally some allowed spoken responses (both corrects and wrongs).

a single file.
The database contains user data, personalizing their experience based on their information and their previous interactions. Each user has an associated ID, and their saved information is: the full name, age, language, schedule, and data about the stress management session.
We use the sqlite3 Python API to define simple functions to interact with the database:

- Add a new user
- Get the next available ID
- Get a list of all the users
- Get the entry of a user by their ID
- Get the entry of a user by their full name
- Update a user’s entry
- Remove a user’s entry

When a user logs-in, their data are loaded on the shared memory using the NAOqi service ALMemory, to limit access to the database to the log-in and eventually, when some data are updated, for example, when the user gives feedback on the stress management session.

4.6 Schedule Interaction

A non-guest user can directly ask the robot to show their schedule. The robot always asks for confirmation, and after that, it starts the MODIM interaction. During that, the user is asked to select between their daily, weekly, or complete schedule. The question is both spoken by the robot and written on the tablet. The user can answer this question either by pressing one of the buttons that appear on the tablet or by pronouncing the desired schedule (Figure 6). The schedule is retrieved by the central memory using the NAOqi module ALMemory and displayed on the tablet. It is visualized as an ordered list of dates with the name of the event associated. The user has plenty of time to inspect their schedule before a timeout will occur (5 minutes), or they can decide to quit the interaction, again by pressing the *quit* button or by pronouncing it.

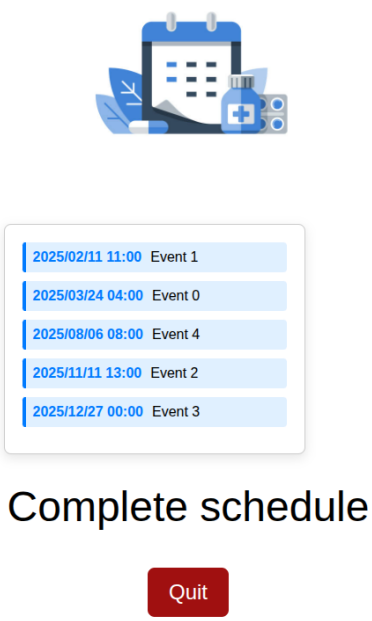


Figure 6: Tablet visualization of the complete schedule requested by a patient

4.7 Stress Management

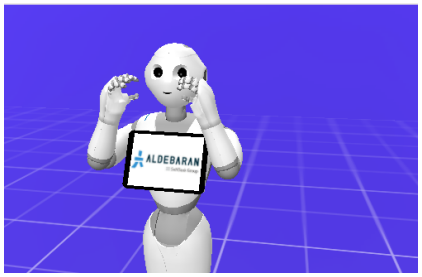
Memory and Adaptation in Stress Management

Memory plays a crucial role in adapting the relaxation exercise to each individual user. Since the goal is to help users feel relaxed, it is essential that the interaction is tailored to be as comfortable and minimally stressful as possible.
Pepper's variability in meeting user needs is managed through five key variables:

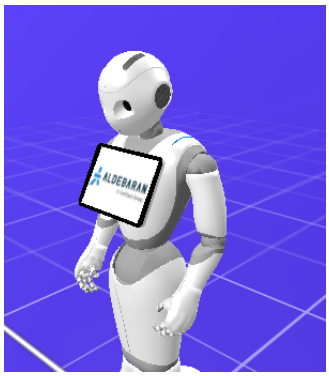
- `last_session`
- `n_cycles`
- `preferred_pace`
- `preference_speed`
- `take_hand`

All these variables are updated during the feedback phase. At the end of the session, the user is explicitly asked if they want to leave feedback, with the question `Would you like to leave a feedback to change some settings?`.
In this first part I will explain more about the feedback, at the end of the interaction, since it will change the start of the Stress management interaction. Within the MODIM action, the first question asked in the feedback is `Did you find the exercise helpful?`, The possible answer options are: `<Very Helpful>`, `<Somewhat>`, `<Not Really>` ,and `<Not At All>`. The first two options will produce the same result for the robot and will be considered as a positive interaction. In this case, Pepper will say: *"I am very happy to have been helpful"*, while raising its arms and moving its torso to express joy (see image 7a and the video to see how the torso moves).

In contrast, for the last two options, which are treated as negative feedback, Pepper will respond: *"I'm sorry I couldn't satisfy you fully. Maybe by making some changes to the exercise, I'm sure it will go better next time, especially with the new modifications"*, while slightly lowering its head(see image 7b).
The next question in the feedback session is about the variable `n_cycles` which represents the number of breathing cycles (inhalation and exhalation) the user performs during the session. The default is 5, but this can be increased or decreased. Each increment or decrement changes the cycle count by 5, providing a balanced progression.
Breathing cycles can be adjusted via `n_cycles` with the question:
"Would you like to modify the number of cycles?"
Possible responses are `increase`, `decrease`, and `no`, modifying the cycle count by 5 per feedback.
`Preferred_pace` and `preference_speed` jointly define the overall rhythm of the interaction. These are adjusted in accordance with the robot's physical constraints (e.g., speech speed ranging from 100 to 400). `preferred_pace` affects the timing of pauses during speech, increasing them for a calmer tone. Although the robot's speaking speed is a configurable parameter, it is currently



(a) Happy expression



(b) Sad expression

Figure 7: Pepper expressing different emotions based on feedback. On 7a upon positive feedback, it shows joy; while on 7b shows sadness after a negative feedback.

commented out in the code due to simulation constraints.
These two variables are modified through the question: *"Would you like a more relaxed tone or would you like me to speak faster?"* with three possible responses, which can be given either by clicking buttons or via voice input: *"Faster," "Slower,"* or *"Neither."* which update `preferred_pace` and `preference_speed` by +100 and +1 respectively.
When the user changes some setting, we make pepper say sentences like *"Great! Next time we will do X cycles"* or *"Ok! Next time I will talk faster."* if the user want a faster rhythm. Each user has an ID to determine if they are new or returning. For new users, the interaction begins differently. For returning users, the `last_session` variable is used to reference previous performance and create continuity, helping the user feel recognized.

Interaction

Now let's talk more about the main interaction and what happens before the last part relative to the feedback.
For new users, the dialogue begins with: *"This is your first session, I will help you throughout the whole experience."* If there was a previous session with a positive outcome, the robot will say: *"I'm sure it will be useful to you again this time. We will do X cycles together."* Otherwise, if the previous session did not go well, Pepper will say: *"I am sorry the last time didn't go as well. Let's try again this time."*
Moreover, if the variables preferred pace and preference speed are higher than a certain threshold (or lower) the robot will add the sentence *"I will go to a faster pace"* (or *"I will go to a slower pace"* After that, the robot will start

the breathing exercise, introducing it with an anchor gesture, which conveys a sense of calm, in order to associate the gesture to the exercise (see 8). After

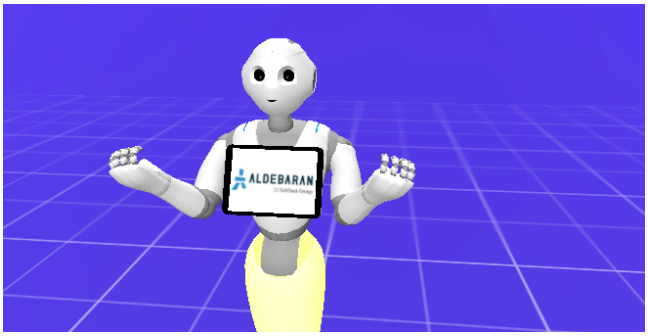
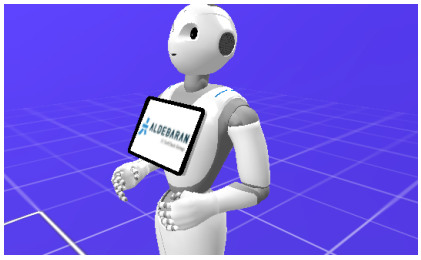


Figure 8: Pepper starts with an anchor gesture the breathing exercise

the gesture, and before starting the breathing cycles, the robot ask if the user want to take its hands. The `take_hand` variable monitors whether the user has accepted or declined hand contact during the last three sessions. If responses are consistent, Pepper acts accordingly in future sessions without repeating the question. If not, the robot will ask again and update the history for the next decision.
This variable is updated based on the question: *"Would you like to hold my hand during the exercise?"* The user can respond with a Yes or No. If the answer is Yes, the robot will say: *"Hold my hand"* and rotate its elbows by 90 degrees (See image 9b).

(a)



(b)

Figure 9: On 9a Pepper asks if user want to take its hands; on 9b Pepper brings his hands forward so that the user can take them.

Just before repeating to breathe in and breathe out, the robot says the two introductory phrases: *"Let's make an effort to feel how our breath acts in the body, what sensations it gives us; let's imagine and perceive fresh air entering through the nostrils or mouth and then coming out warmer."* and *"We concentrate on the movements that the body generates during breathing, and we make a slight effort, but without persisting, to breathe into the belly, we begin to*

feel the breath with the movement of the navel rising and falling, together with the abdomen and chest". Note that Pepper pauses for a variable amount of time between the two sentences, controlled by the previous explained variables, in order to adjust the pace according to the user's preferences. Now the breathing cycle stars, firstly Pepper will say *"If any thoughts or emotions arise, simply notice them and gently bring your focus back to your breath. Let's breathe in together for five seconds"*. As Pepper says this sentence, he leans back and raises his head to simulate inhaling and encourage the user to imitate him(see the video to better understand the motion in section 5). After 5 seconds, Pepper will say *"Now, let's exhale slowly"*, now Pepper will do the opposite movement in order to simulate exhalation. At the end of the breathing cycles, Pepper ends with the phrase *"We've completed our breathing exercises. I hope you feel a little calmer now."* After Pepper will ask for the feedback, as explained before. At the end of the interaction, the robot will ask: *"Would you like to repeat the session, terminate the interaction, or shall I call a nurse to assist you?"* The possible answers are: `repeat`, `terminate`, or `call nurse`. Depending on the user's response, the robot will act accordingly:

- `repeat`: Pepper will say *"Ok, let's start again"* and the session will restart.
- `call`: Pepper will say *"I'm calling a nurse now to ensure you're well taken care of"* and will trigger a nurse call.
- `terminate`: If no feedback-related settings were changed, Pepper will say *"I hope this exercise was useful and that you are feeling better"* and will end the interaction. If instead any of the default parameters have been modified (`N_cycles`, `preferred_volume`, `preferred_pace`, or `preference_speed`), then Pepper will say: *"I'm sure it will go better next time, especially with the new modifications"*.

SOCIAL SIGNAL

Input. In the stress management interaction, Pepper processes multiple social signals:

- **Voice**: for responding to questions and expressing emotions like stress.
- **Tablet Touch**: for confirming choices and giving feedback.

Output. Pepper produces output through various modalities:

- **Tablet Display**: used to show answer options during the interaction 9a or feedback 10 and to emit a full-screen calming blue light during breathing cycles, with minimal information like number of cycles done or whether to inhale or exhale. See figure 11
- **Physical Gestures**: Pepper's body is controlled using angular positioning functions, enabling coordinated motion during exercises and emotional responses. In particular, we utilized Pepper Tools' *setPosture* function that interacts with the ALMotion NAOqi module, which requires a list of

all the values of the angles of the robot joints to be set.
At the beginning of the exercise, Pepper makes an anchor gesture to associate the movement with the type of exercise. If the exercise is finished successfully, the robot raises its hands and moves its torso sideways to express joy. If the session is terminated in a negative manner, Pepper lowers his head. And if the user asks for a hand, pepper rotates the elbow.

- **Sentences**: Phrases that adapt according to the emotions you want to express, such as joy or sadness, and phrases that change according to past experience.

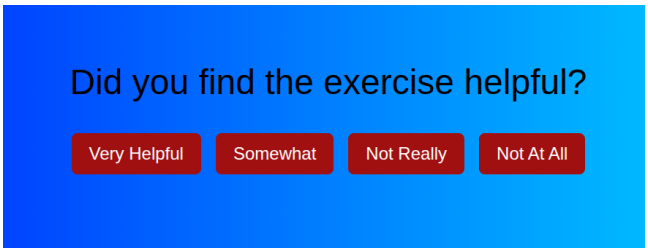


Figure 10: The first question in the feedback part

Figure 11: Monitor the user see during stress management exercise

4.8 Entertainment

Interaction between the user and the Pepper robot can occur through both verbal communication and, specifically during the riddle activity, via the touchscreen tablet located on Pepper's chest (see Section 4.4 for details).
Users can initiate interaction by verbally expressing their preferences regarding the type of activity they would like to engage in. For instance, they may request a riddle or a joke, or they can respond to questions posed by Pepper, both

within the context of the riddle activity and during general conversations aimed at selecting the desired entertainment. This verbal channel allows a natural human-like interaction that supports user engagement and accessibility.
In addition to voice input, users can also interact with the robot through the tablet interface. This secondary input modality is especially relevant during the riddle activity, where the tablet is used to display the current riddle and provide multiple-choice answers; to do it, the user can select its answer by tapping the corresponding on-screen button. This visual and tactile interaction enriches the user experience and provides an intuitive option for users who prefer not to respond verbally.
On the other hand, Pepper communicates through both auditory and visual means. It uses the built-in speakers in its head to speak, telling riddles and jokes, and conversational responses. Simultaneously, during the riddle activity, the tablet presents the riddle text along with five possible answers and supportive visual imagery to enhance the user's understanding and engagement.
Let us now examine this type of interaction in more detail. The activation of the entertainment activity is related to the use of certain expressions by the user.
In the case the user would like to spontaneously initiate an activity that entertains him, he simply asks Pepper to provide him with a riddle or to tell him a joke. To do so, the user has several sentences available to initiate the two activities, like "I would like you to tell me a joke" and "I'd appreciate it if you could offer me a riddle".
On the other hand, Pepper is prepared to act proactively even if the user does not clearly ask him to be entertained; in this case, it will be sufficient for the user to pronounce a sentence such as "I am so bored" for the robot to ask him whether he wants to be entertained with a riddle or a joke. Both the user and Pepper have various phrases to express the same concept, making the interaction more natural and user-friendly.
Five possible answers are provided for the riddles activity; only one of these answers is correct; the chosen answer can be selected by pressing the corresponding button on the tablet or orally. By choosing to answer verbally, the user has various ways to respond, pronouncing the position of the answer (e.g. "third"), the letter or number identifying it (e.g. "A" or "one") or even the answer itself (e.g. "the piano"). After the robot reads the riddle and the consequent appearance of the buttons, the user has a maximum of 60 s to provide the correct answer; if the correct answer is not selected by the end of this time, the message "TIMESUP: you will get it next time!" is displayed. In the event that the wrong answer is given, the user can continue the game until the time limit expires. At the end of this interaction, the message "That's all!" will finally be displayed.
In contrast, joke telling does not involve the use of the tablet interface. These interactions are handled entirely through verbal dialogue, orchestrated via a ".top" dialogue file. This structure allows users to seamlessly request another joke or shift to a different form of entertainment through natural conversation, maintaining the flow and simplicity of the interaction.

5 Results

Along with this report, we presented a video showing some example interactions. The video is available at the following URL: `https://youtu.be/7r7BD27jXx8`. In the video, we show a new patient registering to the robot database, asking for help to stand up and to show his/her schedule. After that, we see a prototypical stress interaction session proposed by the robot after the patient pointed out being stressed. Finally, we show how the robot proposes a riddle for a bored patient and tells a joke.

6 Experimental Evaluation

Research Questions: the study aims to investigate whether replacing the human caregiver with the Pepper robot improves the patient experience and lightens the nurses' workload; in other words, we want to find out whether, in the same care setting, the use of the robot increases patient satisfaction, reduces perceived stress and decreases the subjective workload of the staff.
Research Hypothesis: we formulate three hypotheses. H1: Patients assisted by Pepper show a higher average satisfaction, measured with the Godspeed-Likeability subscale(is part of the larger Godspeed Questionnaire series, used in HRI research to assess how humans perceive robots, in particular it measures how much users like or dislike a robot), than patients assisted by a human caregiver. H2: The introduction of Pepper reduces the subjective workload of nurses, assessed with the NASA-TLX(NASA Task Load Index) score, it is a questionnaire developed by NASA to subjectively measure the perceived workload while performing a task.
H3: The interaction with Pepper alleviates the patients' perceived stress, measured by a questionnaire administered immediately after the session.
Null Hypothesis: H0-1 states that there is no significant difference in patient satisfaction between robotic and traditional care. H0-2 states that the average NASA-TLX score of nurses remains unchanged with or without Pepper. H0-3 states that the average level of stress perceived by patients does not change according to the type of care received.
Experimental Variables: The independent variable is the type of assistance, distinguished as either robotic with Pepper or traditional with a human operator. The dependent variables cover the three pillars of the HRI evaluation: user satisfaction measured with the Godspeed-Likeability subscale, post-interaction stress measured with a standard questionnaire, effectiveness expressed by the percentage of requests completed without external intervention and, for staff, workload measured with the NASA-TLX supplemented by the average heart rate recorded during the shift.

6.1 Evaluation Protocol

6.1.1 Aim of the Study

Check whether the use of *Pepper* as a substitute for traditional assistance does:

- increases the **satisfaction** of patients (Godspeed–Likeability scale);
- reduces their **perceived stress** (post-session questionnaire);
- lowers the subjective **workload** of nurses (NASA-TLX).

We consider the **Within-Subject** experimentation, it requires fewer participants, since the same patient tests both conditions. This makes it easier to test. Since we want to compare the two types of care (human vs. robotic), half of the patients start with the nurse and then switch to Pepper; the other half follow the reverse order.
Non-critical, voluntary patients, able to understand simple instructions, are analysed.

6.2 Scenario and Tasks

1. **First interaction** with the assistant assigned first (nurse or Pepper).
2. Compilation of Godspeed–Likeability and stress questionnaire.
3. **Pause/Wash-out**.
4. **Second interaction** with the other type of assistance.
5. Second filling out of the same questionnaires.

The nurses complete the NASA-TLX at the end of the shift.

6.3 Data Collected

- **User Satisfaction**: Godspeed–Likeability score.
- **Perceived Stress**: short Likert scale post-interaction.
- **Nurse Workload**: NASA-TLX and average heart rate.
- **Effectiveness**: "task completed unaided" flag noted by an observer.

An identical operating script is considered for both trials, along with the presence of trained observers and short questionnaires, to minimise fatigue.

7 Conclusion

This project shows how human-robot interactions do not focus only on accomplishing a specific task but also on showing social-cognitive skills. In a healthcare facility, a robot assistant needs to show empathy and act accordingly to the human's current state to facilitate the recovery and reduce the stress load. To demonstrate social skills, the robot utilizes various communication modalities, including movements, speech, and tablet interactions, and adapts to the user's situation based on the feedback received and past experiences.

A more sophisticated solution can utilize visual data to enhance social-cognitive skills. A face recognition system allows for easier user identification, thus making the initiation of each interaction more natural. Visual cues, along with an emotion recognition system, can be exploited to adapt the robot's behaviour to the perceived feelings of the patient. Finally, a navigation module can be integrated, so as to include movements that involve more dynamic interactions that do not assume the robot is still in front of the user.

Regarding non-technical considerations, several important issues need attention. From an ethical and societal standpoint, responsibility remains a critical concern: in case of errors or malfunctions, who bears accountability? Clarifying liability is essential for maintaining trust and safety. Furthermore, The psychological dimension raises questions about acceptability: are patients comfortable being assisted by robots instead of human caregivers, and will they be willing to embrace such technology?

From an economic perspective, deploying advanced robotic technology like Pepper entails significant initial investments. Institutions must evaluate the long-term cost-effectiveness compared to traditional approaches. Finally, from an organizational viewpoint, introducing robotics into hospital environments requires redefining professional roles and responsibilities, along with providing healthcare staff adequate training to ensure smooth integration and effective utilization of these new tools.

Addressing these non-technical aspects is fundamental to successfully incorporating robotic solutions into healthcare systems, ensuring alignment with ethical standards, societal acceptance, economic viability, and effective organizational adaptation.

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